# EVRYN Evidence Grounding Contract: Dual-Span Provenance & Causal Directionality

**Cause Entity (C)**

Canonical: PmrB

Mention: "PmrB sensor kinase"

Entity Type: Sensor Kinase / Protein

Section: Results | Page: 2

**RELATION: DIRECT_CAUSE (+)**

**Direction: FORWARD | Confidence: 0.95 | Polarity: POSITIVE**

**Effect Entity (E)**

Canonical: PmrA

Mention: "response regulator PmrA"

Entity Type: Response Regulator / Protein

Section: Results | Page: 3

**Primary Evidence Sentence & Verbatim Provenance Coordinates (S_causal)** — **100% PROVENANCE VERIFIED**

*"Acid pH activates the PmrA/PmrB system by promoting PmrB-mediated phosphorylation of PmrA."*

**Doc ID: PMC1804205 | Section: RESULTS | Paragraph: 14 | Sentence ID: s42 | Offsets: [1420--1512] | Page: 2**

**Intra-Sentence Context**

**1. Direct Intra-Sentence Assertion** Single Sentence

- Both C & E co-occur in S_causal boundary
- Syntactic trigger: "activation of", "induces"
- Complete coordinates: <p.2, Results, [1420-1512]>
- High empirical precision (0.90--0.98 on gold)

**Anaphoric Context Resolution**

**2. Context-Resolved Candidate** Cross-Sentence

- Inter-sentence alignment across [S_k, S_{k+1}]
- Demonstrative pronoun & anaphora resolution
- Dual Provenance Binding: [S_cause, S_effect]
- Abbreviation expansion via domain normalizer

**Transitive Multi-Hop Linking**

**3. Multi-Page Transitive Link** Multi-Page Graph

- Path traversal: A -> B (p.2) -> C (p.8)
- Inter-page distance: Delta_p >= 1, hops <= 2
- Pruned by domain invariants (TCSTS_REVERSE_PAIRS)
- Distance decay: conf = prod(c_i) * lambda^d